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MILITARY SPECIFICATION

FUSE, CURRENT LIMITER TYPE, AIRCRAFT.

This specification has been approved by the Department of the Air Force and by the Bureau of Naval Weapons.

1. SCOPE

1.1 Scope.- This specification covers the requirements for current limiter type fuses for use in aircraft primary electrical power systems conforming to MIL-STD-704. These limiters are suitable for operation at -54° to +125° C.

1.2 Classification.- Limiters shall be as described in the applicable Military Standard (MS). (See 6.2.)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-F-5373	Fuseholders, Block Type, Aircraft
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MS20659	Terminal, Lug, Crimp Style, Copper, Uninsulated, Class I
MS25190	Wire, Electric, 600 and 3000 Volt, Copper, Aircraft, 105° C

FSC 5920

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification.- The current limiters furnished under this specification shall be a product which has been subjected to and which has passed the qualification tests specified herein, and which has been listed on or approved for listing on the applicable Qualified Products List (see 6.3).

3.2 Unless otherwise specified in the contract or order, no data (other than reports and drawings accompanying qualification samples) are required by this specification or any of the documents referenced in section 2, herein (see 6.2).

3.3 Materials.- Materials shall conform to the applicable MS and as specified herein. Materials not covered by an applicable MS or not specifically described herein shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended. Plastic materials which are non-nutrient to fungus shall be used in construction of the limiter.

3.3.1 Metals.- All metals used in construction of limiters shall be corrosion resistant or suitably protected to resist corrosion. Use of dissimilar metals, especially contacts between brass, copper, or steel and aluminum or magnesium alloys, shall be avoided. Where contact between dissimilar metals is unavoidable, the metals shall be protected against electrolytic corrosion.

3.3.2 Plastic materials.- Plastic materials used in the limiter shall be as specified on the applicable MS. Plastic materials shall neither support combustion nor give off noxious gas when subjected to arcs, such as those caused by interrupting heavy short-circuit currents or explosions of gaseous vapors. Plastic materials with cellulose fillers shall not be permitted.

3.3.2.1 Color.- The color of external plastic parts shall be as specified on the applicable MS. After exposure to the temperature conditions specified in 4.6.3 there shall be no degradation of these requirements.

3.3.2.2 Protective treatment.- Protective coating shall not be used.

3.4 Design and construction.- The current limiter shall conform to the requirements specified in the applicable MS and herein.

3.4.1 Visual indication.- A blown limiter shall be self-indicating and so designed that visual indication of an open condition will not require removal from the limiter holder. The design shall be such that an energized electrical circuit will not be necessary to visually indicate an open-circuit limiter after a fault has occurred.

3.4.2 Insertion and removal.- Except for bolt-down types, the design shall be such that the limiter can be easily inserted into or removed from the limiter holder without requiring use of a tool. The contour of the molding shall provide finger grips or other means to facilitate removal by hand.

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3.4.3 The limiter shall be designed for use with holders in accordance with MIL-F-5373.

3.4.4 Mounting position.- Limiter performance shall not be affected by its mounting position.

3.4.5 Terminals.- Limiter terminals shall be as specified on the applicable MS.

3.4.6 Current rating.- The rating of the limiter shall be as specified on the applicable MS.

3.4.7 Dimensions.- The dimension of the limiter shall be as specified on the applicable MS.

3.4.8 Weight.- The weight of the limiter shall be as specified on the applicable MS.

3.5 Performance.- Limiters shall pass the performance requirements of this specification and the applicable MS when tested as specified in section 4.

3.5.1 Time-current characteristics.- Time-current characteristics shall be as specified on the applicable MS.

3.5.2 Initial loading.- Loads up to 100 percent of rating may not reduce by more than 10 percent the current required to melt the limiter in any given time (see 4.6.11).

3.5.3 Ambient temperature influence.- The time-current characteristics of the limiter shall be within the tolerance limits specified on the applicable MS (see 4.6.3).

3.5.4 Overloading.- The characteristics specified herein shall not be altered by overloading the limiter up to 70 percent of the current required to melt the limiter in a specified time, as shown on the applicable MS (see 4.6.10).

3.5.5 Terminal voltage drop.- The voltage drop across the limiter terminals, while the limiter is carrying rated current, shall not be greater than 0.1 volt (see 4.6.20).

3.5.6 Alternating current interrupting capacity.- When specified, the current limiter shall be capable of interrupting prospective line-to-neutral faults on a 115/200 volt (V), three phase, 400 cycles per second (cps) system that is capable of producing a line-to-neutral fault current of 6,000 amperes (amps) during the first cycle of fault current; for a sustained fault, the current may be allowed to decrease at a gradual rate to 3,600 amps within 0.05 second after fault initiation. Two limiters, connected in series, shall be capable of interrupting line-to-line faults on a system conforming to MIL-STD-704 and capable of producing a line-to-line fault current of 4,200 amps during the first cycle of fault current and, for a sustained fault, the current may be allowed to decrease in a gradual rate of 2,500 amps within 0.05 second after fault initiation.

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3.5.7 Direct current interruption capacity.- When specified, the limiter shall be capable of interrupting prospective fault currents of 6,000 amps, with this value achieved in 0.01 to 0.03 second after fault initiation. The system voltage shall be not less than 28V before the fault. The transient recovery voltage at the terminals of the limiter holder shall be not less than 28V within 0.002 second after the interruption and shall maintain 28V at maximum value with the steady-state, open-circuit voltage reaching no less than 28V after the fault.

3.5.8 Temperature rise.- The temperature rise shall not exceed 90° C when subjected to the test of 4.6.21.

3.5.9 Environment.- The limiter shall perform satisfactorily when subjected to any natural combination of the following environmental conditions. There shall be no breakage, malfunction, or evidence of damage that would prevent additional testing.

3.5.9.1 Temperature.- Temperatures shall be from -54° C to +125° C (see 4.6.2 through 4.6.5).

3.5.9.2 Altitude.- Pressure altitude shall be from sea level to 60,000 feet (see 4.6.14 through 4.6.16).

3.5.9.3 Moisture resistance.- Moisture resistance shall be as specified in 4.6.4.

3.5.9.4 Salt spray.- There shall be no peeling, chipping, or blistering of metal or plastic surfaces, or exposure of base metal following exposure of the limiter to the test of 4.6.7.

3.5.9.5 Sand and dust.- Exposure to sand and dust shall be as specified in 4.6.17.

3.5.9.6 Vibration.- Following testing in accordance with 4.6.12, there shall be no physical or electrical damage to the limiter and each limiter shall meet the room ambient time-current requirements of the applicable MS at 500 percent of rated current (see figure 1).

3.5.9.7 Mechanical shock.- Mechanical shock shall be as specified in 4.6.19. Following this test, the limiter shall withstand the time-current test of 4.6.13 for 500 percent of the rated current.

3.5.9.8 Indicator melt time.- The indicator melt time shall be as specified in 4.6.6.

3.5.9.9 Acceleration.- Acceleration shall be as specified in 4.6.18.

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3.5.9.10 Explosion proof.- Operation shall be as specified in 4.6.16.

3.5.10 Terminal strength.- There shall be no physical damage to the limiter following application of the forces specified in 4.6.9. This requirement shall not apply to bolt-down type limiter fuses.

3.5.11 Thermal shock.- The limiter shall withstand the thermal shock test of 4.6.5 with no evidence of limiter crackage, delamination of finishes, obscuring of identification markings, severance of the fuse element, loosening of parts, or false indication. Following this test, the limiter shall withstand the time-current test of 4.6.13 for 500 percent of the rated current.

3.5.12 Random drop.- There shall be no physical damage and the limiter shall remain in proper position with respect to the fuseholder clips when subjected to the random drop test of 4.6.8.

3.5.13 Life test.- There shall be no physical damage to the limiter as a result of the life test specified in 4.6.2.

3.5.14 Rated current.- Subsequent to the tests specified in 4.6.22 the limiter shall be capable of passing rated current for 1 minute.

3.6 Identification of product.- The limiter shall be marked for identification in accordance with MIL-STD-130. The marking shall include the manufacturer's name or symbol, current rating of the limiter, and applicable MS part number.

3.7 Workmanship.- The limiter shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of marking of parts and assemblies, welding and brazing, and freedom of parts from burrs and sharp edges. No part shall work loose in service.

3.7.1 Cleaning.- The limiter shall be thoroughly cleaned of metal chips and other foreign material, before and after final assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests.- The inspection and testing of limiters shall be classified as:

(a) Qualification tests (4.3)
(b) Quality conformance tests (4.4)

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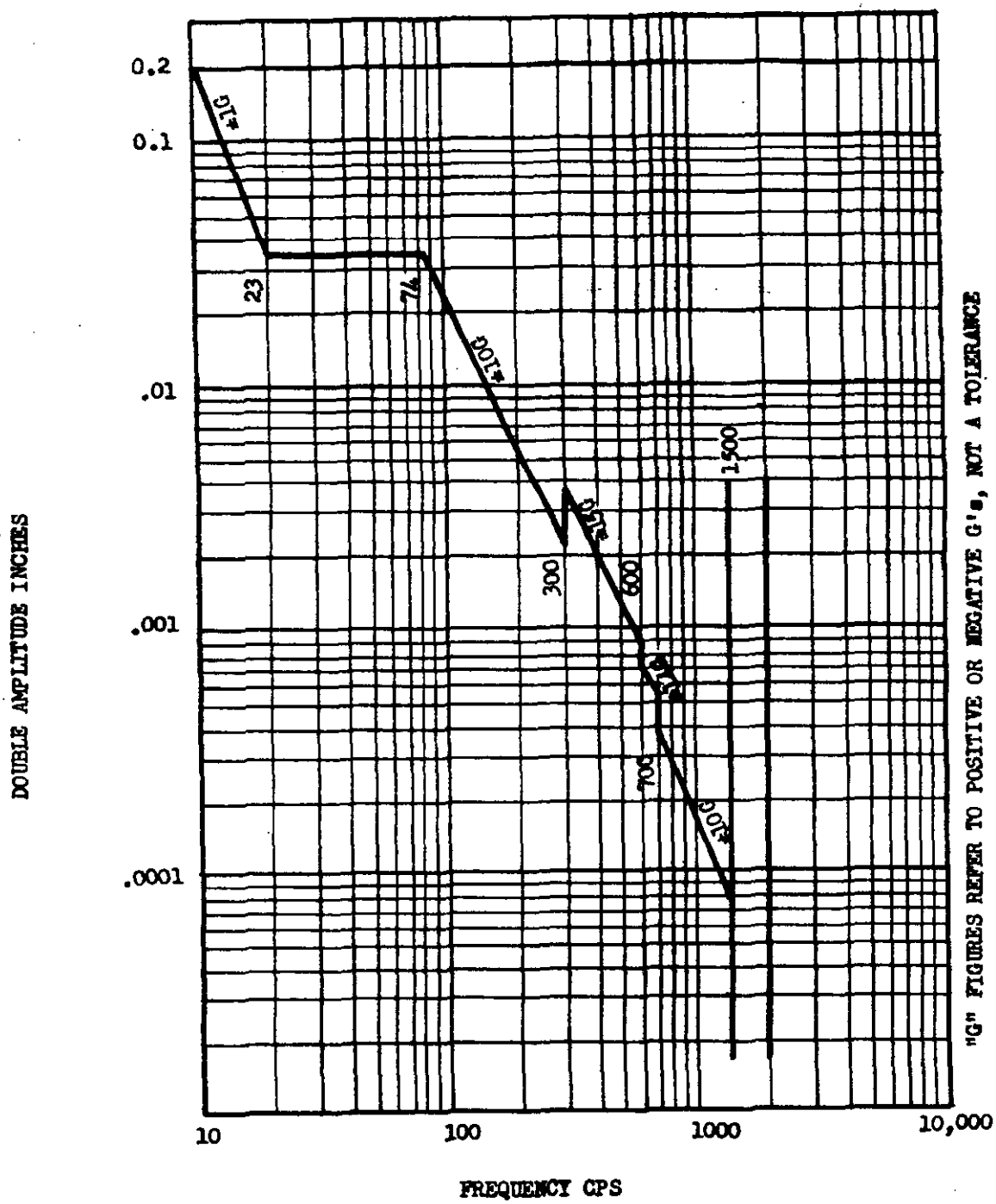


FIGURE 1. Vibration curve.

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4.3 Qualification tests.- Qualification tests for fuses of a single rating shall consist of all the tests specified in 4.6. For group qualification of fuses with the same physical sizes and construction, the maximum and minimum ratings shall be subjected to all the tests specified in 4.6, and all intermediate ratings shall be subjected to the tests for time-current, overload, and shock. All qualification tests shall be performed in the order shown in table I.

4.3.1 Sampling instructions.- Qualification test samples shall consist of not less than 57 limiters of each current rating. Samples shall be representative of the manufacturer's normal production and shall be identified by the manufacturer's part number.

4.3.1.1 Test samples.- Test samples shall be submitted for qualification testing as designated by the activity responsible for qualification (see 6.3). Certified test data covering nondestructive tests for the type limiter for which approval is sought shall be submitted to the activity responsible for qualification. This report shall be submitted prior to qualification testing. Samples shall be plainly identified by securely attached durable tags marked with the following information:

Sample for qualification test
FUSE, CURRENT LIMITER TYPE, AIRCRAFT
Name of manufacturer
Limiter designation military standard and dash number
Submitted by (name) (date) for qualification tests in accordance with MIL-F-5372, under authorization (reference authorizing letter).

4.4 Quality conformance tests.- Quality conformance tests shall consist of:

(a) Individual tests (4.4.1)
(b) Sampling tests (4.4.2)

4.4.1 Individual tests.- Each limiter shall be subjected to an examination of product as specified in 4.6.1.

4.4.2 Sampling tests.- From each lot of current limiters manufactured under substantially the same conditions and at essentially the same time, 3 samples selected in accordance with the sample size indicated in table II shall be selected at random and tested in accordance with the test requirements of 4.6.13, using 240, 400, and 1,000 percent of current rating only, respectively, for each sample. The results for each individual current limiter shall be within the allowable variation obtained from the time-current characteristic curve of the applicable MS. If an individual test result exceeds the allowable percent variation, the lot will not be acceptable. Time-current results shall be recorded as plus or minus percentage of variation. If all results are within the allowable variation, calculate the average, $\bar{X}$, and the standard deviation, D, of the percent of variation from the characteristic curve. If the $\bar{X}$ + KD is equal to or less than the plus percent variation (T_u) from the time-current curve as specified in the applicable MS, and $\bar{X}$ - KD is equal to or greater than the minus percent variation (T_l), the lot will be acceptable, in accordance with table II. Also, in accordance with table II, if $\bar{X}$ + KD is greater than the plus percent variation (T_u), or if $\bar{X}$ - KD is less than the minus percent variation (T_l), the limiters shall be rejected.

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TABLE I. Qualification test schedule current limiters

	Requirement para.	Qualification test	Test para.	Test sample numbers											
				1-2	3-8	9-10	11-12	13-14	15-16	17-18	19-20	21-22	23-32	33-40	41-42
	4.6.1	Exam. of product	4.6.1	1	1	1	1	1	1	1	1	1	1	1	1
	3.5.13	Life test	4.6.2	2											
	3.5.3	Amb. temp. inf.	4.6.3		2										
1/	3.5.9.3	Moisture resis.	4.6.4			2									
1/	3.5.11	Thermal shock	4.6.5				2								
1/	3.5.9.4	Salt spray	4.6.7					3							
	3.5.12	Random drop	4.6.8						2						
1/	3.5.10	Terminal strength	4.6.9					2							
	3.5.4	Overload damage	4.6.10							2					
	3.5.2	Initial loading	4.6.11								2				
	3.5.9.6	Vibration	4.6.12									2			
	3.5.1	Time-current	4.6.13	3	3	3	3					3	2		
	3.5.6	Ac rupture	4.6.14											2	
	3.5.7	Dc rupture	4.6.15												2
	3.5.14	Rated current test	4.6.22					4	3						

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TABLE I. Qualification test schedule current limiters (Cont'd.)

	Requirement para.	Qualification test	Test para.	Test sample numbers				
				43-49	50-51	52-53	54-55	56-57
	4.6.1	Exam. of product	4.6.1	1	1	1	1	1
1/	3.5.9.10	Explosion proof	4.6.16	2				
1/	3.5.9.5	Sand and dust	4.6.17		2			
	3.5.9.9	Acceleration	4.6.18			2		
	3.5.9.7	Mech. shock	4.6.19				2	
1/	3.5.5	Voltage drop	4.6.20					2
1/	3.5.8	Temp. rise	4.6.21					3
	3.5.14	Rated current test	4.6.22					4
	3.5.1	Time-current	4.6.13		3	3	3	

The test sequence is determined by the number under the test sample numbers. 1 = first; 2 = second; 3 = third; 4 = fourth.

1/ These tests not to be performed on those limiter fuses that are inactive for new design provided that those limiter fuses are procured from an otherwise qualified source.

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TABLE II. Sampling plan

Plan lot size	Random sample quantities	Sample factor K	Accept if	Reject if
0 - 1,300	6	2.225	$\bar{X} + KD \leq T_u$ and $\bar{X} - KD \geq -T_l$	$\bar{X} + KD > T_u$ or $\bar{X} - KD < T_l$
1,301 and over	15	2.195	$\bar{X} + KD \leq T_u$ and $\bar{X} - KD \geq -T_l$	$\bar{X} + KD > T_u$ or $\bar{X} - KD < T_l$

Definition of Symbols:

$\bar{X}$ = Average results.
X = Percent deviation of fusing current from mean curve at a point on the curve.
n = Sample size (the number of limiters selected at random from the lot for testing).
K = Sample factor.
D = Standard deviation of the percent variation from the characteristic curve.

$$D = \sqrt{\frac{\Sigma X^2 - \frac{(\Sigma X)^2}{n}}{n-1}}$$

T = Tolerance, allowable fusing current variation from characteristic curve, obtainable from applicable detail specification.
T_u = Upper allowable tolerance.
T_l = Lower allowable tolerance.

Example computation:

Sample number	Percent current variation from characteristic curve (X)	X^2
1	+ 7.0	49.0
2	- 2.0	4.0
3	+ 3.5	12.25
4	- 3.0	9.0
5	- 1.0	1.0
6	+ 6.0	36.0
	$\Sigma X = 10.5$	$\Sigma X^2 = 111.25$

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Since all results (X) are within the allowable ±8 percent variation, acceptance of the lot is dependent upon the following calculations for $\bar{X} \pm KD$:

$$D = \sqrt{\frac{111.25 - \frac{(10.5)^2}{6}}{5}} = \sqrt{18.575} = 4.31 \text{ percent}$$

$$\bar{X} = \frac{\Sigma X}{n} = \frac{10.5}{6} = 1.75$$

$$\bar{X} + KD = 1.75 + (2.225)(4.31) = 11.33 \text{ percent}$$

$$\bar{X} - KD = 1.75 - (2.225)(4.31) = 7.38 \text{ percent}$$

Since $\bar{X} + KD = 11.33$ percent $> T_u = 8$ percent, reject the lot.

4.4.2.1 Special attribute sampling provisions.- In using a variable plan, it is possible to reach a reject decision without actually having observed any defective items in the sample. In such instances, the user may, if he desires, select sufficient additional sample units as required to satisfy the requirements for attribute inspection in accordance with MIL-STD-105, employing an AQL equal to 1.0 percent defective sampling plan. In the event that an acceptance decision is reached under the attribute inspection, the lot shall then be considred as having fulfilled the acceptance inspection requirements.

4.4.2.2 Lot.- The lot definition, information, and size shall be in accordance with MIL-STD-105.

4.4.2.3 Continuance of individual tests.- For production reasons, individual tests may be continued pending investigation of a sampling test failure, but final acceptance of the entire lot shall not be made until it is determined that the lot meets all the requirements of this specification.

4.4.2.4 Defects in items already accepted.- The investigation of a test failure could indicate that defects may exist in items already accepted. If this occurs, the contractor shall advise the procuring activity of all defects likely to be found and methods of correcting them.

4.5 Test conditions.-

4.5.1 Standard conditions.- Unless otherwise specified, all load tests shall be conducted in still air at a temperature of 25 ± 3° C which shall be considered room ambient and at sea level pressure. The atmospheric pressure at the test facility may be used in lieu of sea level pressure if the elevation of the test facility is not greater than 3,000 feet above sea level.

4.5.2 Power supply.- Unless otherwise specified, the power source for all ac current tests shall be 118 ±3V at a frequency of 400 ±20 cps with waveform and harmonic content in accordance with the requirements of MIL-STD-704, and for dc shall be 28 ±1 volts.

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4.5.3 Test cables.- Test cables in accordance with table III shall be used for all tests that require limiters to carry current except the tests requiring currents of fault magnitude in the rupture and explosion proof tests. Test leads from the fuseholder to the source of power shall be a minimum of three feet in length and shall conform to MS25190.

TABLE III. Test leads

Limiter rating (amperes)	Wire size	Terminal lug MS part No.
1 thru 10	20	MS20659-2
15	18	MS20659-2
25 - 30	14	MS20659-4 or -27
35 - 40	12	MS20659-5 or -28
50	10	MS20659-5 or -28
60	8	MS20659-7 or -29

4.5.4 Test equipment and inspection facilities.- Test equipment and inspection facilities shall be of sufficient accuracy, quality, and quantity to permit performance of the required acceptance inspection. The manufacturer shall establish adequate calibration of test equipment to the satisfaction of the Government.

4.5.5 Limiter holders.- All tests requiring holders shall be performed with applicable holders as specified in MIL-F-5373.

4.6 Test methods.-

4.6.1 Examination of product.- All limiters shall be inspected to verify that materials, design, construction, weight, physical dimensions, marking and workmanship conform to applicable requirements (see 3.3 through 3.4.8, 3.6, and 3.7).

4.6.2 Life test.- The limiter shall be inserted in the specified limiter holder and subjected to a 1,000-hour life test consiting of 500 hours of operation at each of the ambient temperature extremes of -54° ±5° C and 125° ±5° C. While at each of these temperature extremes, the limiter shall carry rated current continuously for 250 hours and then be subjected to 2,500 off and on cycles with each cycle consisting of 3 minutes at rated current and 3 minutes at no current. The temperature of the limiter bodies shall be monitored continuously and the temperature rise determined and recorded. The maximum temperature shall not exceed 218° C. Time-current measurements performed subsequent to this test shall be distributed evenly at 500 and 1,000 percent of current rating. The time-current test method shall be in accordance with 4.6.13.

4.6.3 Ambient temperature influence.- Fifty percent of the limiters, as required for this test per table I, shall be conditioned at the low ambient temperature extreme of -54° ±5° C and the remaining 50 percent shall be conditioned at the high ambient temperature extreme of 125° ±5° C. The limiters shall be stabilized at each temperature extreme for a minimum of 4 hours. After thermal stabilization at these temperature extremes, the time-current measurements of the limiters shall be obtained at 500 and 1,000 percent of the current rating. These measurements shall

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be in accordance with the requirements of 4.6.13 except at the temperatures stated in this paragraph. The specified limiter holder with at least a 3-foot length of wire attached to the limiter holder terminals shall be stabilized at the same ambient temperature as the limiters.

4.6.4 Moisture resistance.- The limiter shall be inserted in the specified limiter holder and subjected to the moisture resistance test, Method 106B of MIL-STD-202, except that step 7B of this method shall be omitted. Subsequent to this test and immediately after a 24-hour conditioning at 25° ±2° C and at a relative humidity of 50 ±5 percent, the limiter shall be tested and shall meet the time-current requirements of 4.6.13 for 500 percent of rated current.

4.6.5 Thermal shock.- The limiter shall be inserted in the specified limiter holder and subjected to 5 thermal shock cycles in accordance with Method 107B, test condition B of MIL-STD-202, except that -54° C shall be used in lieu of -65° C (see 3.5.11).

4.6.6 Indicator melt time.- The indicator melt time shall be determined as indicated from the oscillograph shown on figure 2 and shall not exceed 25 percent of the melt time at the 1,000 percent of rated current test.

4.6.7 Salt spray.- The limiter shall be inserted in the specified limiter holder and subjected to 48 hours of salt spray exposure in accordance with Method 101B of MIL-STD-202, using a salt solution concentration of 5 percent.

4.6.8 Random drop.- The limiter shall be inserted in the specified limiter holder which shall be rigidly mounted in a mounting fixture and subjected to the random drop test in accordance with Method 203A of MIL-STD-202. The limiters shall remain inserted in the limiter holders throughout the test.

4.6.9 Terminal strength test.- The limiter shall be clamped firmly at one end and the forces specified in table IV applied to the opposite end. This test does not apply to bolt-down type limiter fuses.

TABLE IV. Terminal strength test

Direction of application of force	Ampere ratings	
	Up to 30	Above 30
	Force in pounds (minimum)	
To edge of blade, as inserted into clip of holder	10	25
To flat side of blade	5	10
To end of blade so as to push into limiter body	30	40

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4.6.10 Overload damage.- Using equipment suitable for the time-current tests of 4.6.13, with provisions added to interrupt current at predetermined times before limiter damage, apply the test current for the time shown on the applicable time-current curve for 70 percent of the melting time at 240,300, 500, and 1,000 percent of rated current. With the limiter stabilized at the ambient temperature of 125° ±5° C, apply each of the test currents one time, allowing a minimum of 10 minutes between applications. Stabilize the limiter at room temperature and perform the calibration test of 1,000 percent of rating. Failure will be indicated by inability of the limiter to complete the above procedure or to melt at a time outside the calibration requirement of 4.6.13.

4.6.11 Initial loading influence.- Adjust the limiter to 100 percent of rated current. Maintain this current until temperature stabilization is achieved at the limiter holder clips. While at this condition, apply the 1,000 percent calibration current, and measure the fuse melt time by means of an oscillograph.

4.6.12 Vibration.- The vibration test shall be accomplished in accordance with Procedure XII of MIL-E-5272 with the following additions and exceptions:

(a) The vibration curve, figure 1 of this specification, shall be used for the frequency scan, resonance endurance, and cycling endurance tests.

(b) Three similar samples of the limiter shall be inserted in the applicable MS holder and mounted on the vibration fixture. The center limiter and one other shall carry 100 percent of rated current at 400 cps while the third limiter shall carry no current for the entire vibration test.

(c) The limiters and limiter holder shall be subjected to the test ambient temperature for at least 30 minutes prior to the vibration test. The limiters that carry current shall have current applied for at least 30 minutes prior to the vibration test. The following schedule defines the temperatures and times to be used:

Procedure	Ambient temperature		
	25° C	125° C	-54° C
	Perform	Perform	Perform
Frequency scanning	0.667×10^6 cycles, for no more than 5.33 hours	0.167×10^6 cycles, for no more than 1.67 hours	0.167×10^6 cycles, for no more than 1.67 hours
Resonance endurance	1.5 hours	———	———
Cycling endurance			

(d) Suitable test circuitry with indicating provisions shall be connected to the terminals of the limiter holder to establish that electrical continuity is maintained by each limiter during the entire vibration test.

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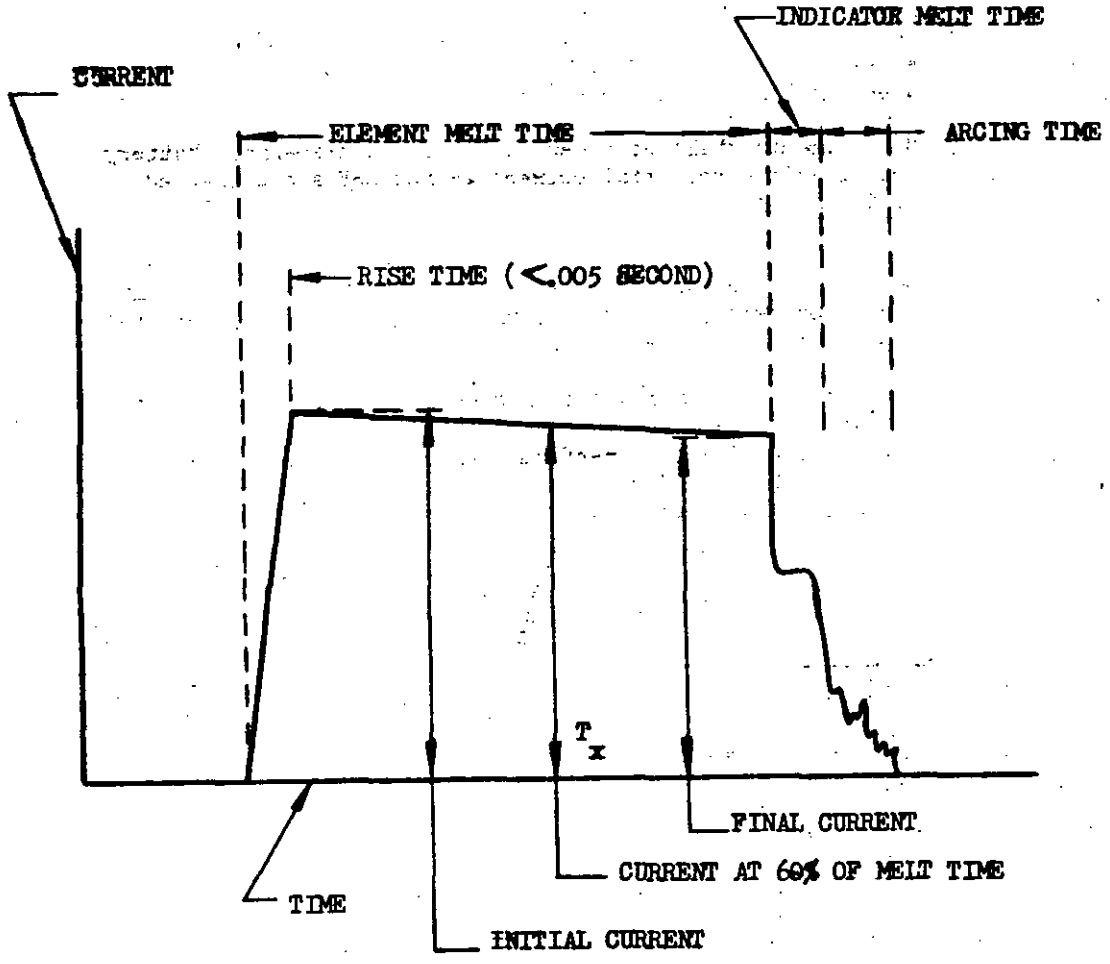


FIGURE 2. Typical time current oscillograph.

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4.6.13 Time-current measurements.- Time-current measurements shall be:

(a) Time-current measurements shall be taken at an ambient temperature of 25° ±2° C on 2 samples of each rating of each type of current limiter at 200, 240, 300, 500, and 1,000 percent limiter rating.

(b) For each test current, the load bank shall be adjusted to a value that will result in ±5 percent of the current specified when the limiter is blown (melted).

(c) Measurements shall be taken, utilizing a high-drain battery source with a potential between 28 and 48V and connected similar to figure 3.

(d) An oscillograph record shall be made of time-current measurements of 10 seconds or less. The limiter element melt time, indicator melt time, and fusing current shall be determined from these records and tabulated. (See figure 2.)

(e) The percent deviation of fusing current from the curve of the applicable MS shall be calculated and recorded. The measurements shall be performed with the limiters inserted in the limiter holders as specified in the applicable MS. Percent current deviation (X) is defined as follows:

$\%$ deviation $= X = (-100)\ \frac{I_c - I_x}{I_c}$; where I_x is the current that melts the test limiter at the time t_x (see figure 2) and I_c is the current obtained from applicable time-current curve at time t_x.

(f) The limiter shall conform to the tolerance stated on the applicable time-current curve.

(g) The limiter shall not suffer external physical damage as a result of this test.

4.6.13.1 Test apparatus.- The test apparatus shall be:

(a) Load current meter, accurate to ±0.5 percent.

(b) Recording oscillograph and galvanometer, complete with shunts and amplifier, shall have light beam galvanometers with a minimum linear frequency response of 3,000 cps; shall have paper speed, paper width, and a timing marker to allow time measurements accurate to the nearest 0.5 millisecond; and current and voltage measurements accurate to ±2 percent.

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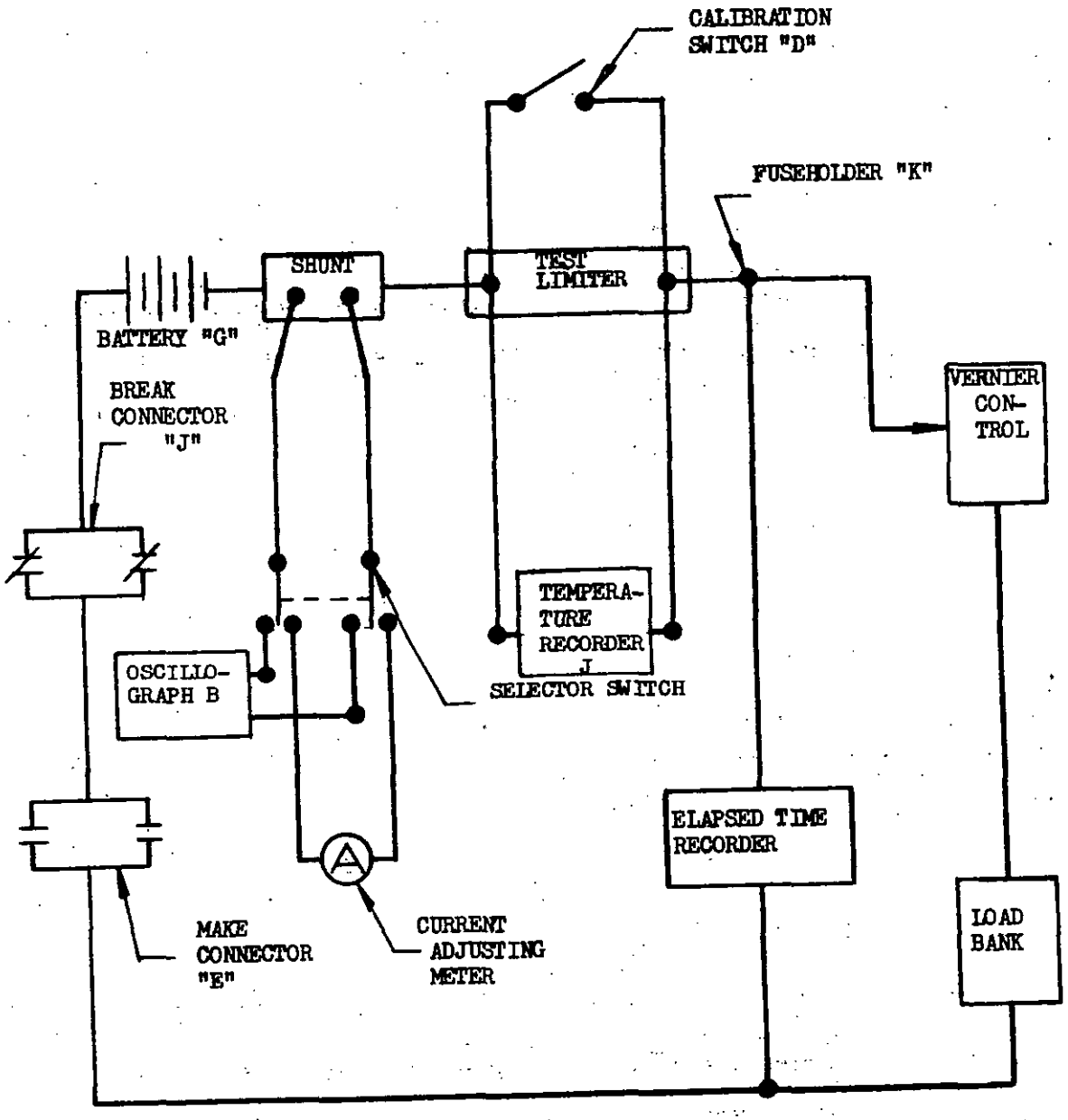


FIGURE 3. Time-current schematic.

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(c) Load bank resistive, capable of setting current flow to desired value ±5 percent, current droop not to exceed 10 percent; provide cooling if necessary.

(d) Switch, calibration, to be utilized for presetting desired current flow.

(e) Contactor, make, to be utilized for making current only; the contact bounce shall be negligible, less than 250 microseconds.

(f) Electric stop clock, calibrated to 0.01 second, accurate to ±0.02 second, to be used for measurements exceeding 10 seconds.

(g) Battery, 28 to 48V, capable of supplying the necessary test currents for the required times with current droop limited to 10 percent and rise time to 0.005 second maximum.

(h) Cable, No. 4/0, capable of passing the highest test current with no appreciable voltage drop.

(i) Test enclosure, free of air drafts and capable of maintaining samples at an ambient temperature of 25° ±2° C.

(j) Potentiometer, to monitor temperature of limiter clips.

(k) Limiter holder, as specified in MIL-F-5373.

4.6.13.2 Fuse element melt time, less than 10 seconds.- On tests of less than 10 seconds element melt time, the time for the current to reach steady-state condition (rise time) shall be less than 0.005 second. The short time to reach steady-state conditions shall be achieved by using high-drain batteries and short leads to reduce the transient impedance of the circuit. All leads shall be bundled as close as possible, and shall be isolated from magnetic material. The leads shall not run in magnetic conduit. Current droop caused by the increase in limiter resistance, as the limiter temperature increases from ambient temperature to the melt temperature, shall not exceed 10 percent of the initial current value.

4.6.13.3 Measurement of limiter element melt time.- When determining melt time of a limiter element from oscillographic records, time shall be measured along a line parallel to the zero trace to the point where the current drops sharply. The sharp current drop may be caused by the indicator or the arcing element. Fusing current values shall be measured (I_x) at a point where 60 percent of the melt time has elapsed. Figure 2 depicts a typical oscillographic current trace.

4.6.13.4 Measurements greater than 10 seconds.- For element melt time greater than 10 seconds, an electric stop clock may be used. The fusing current may be maintained at the correct value by varying the setting of the rheostat. In melt time exceeding 10 seconds, the indicator melt time and the arcing time become insignificant in comparison with the total interrupt time, therefore, it will be permissible to discount them.

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4.6.13.5 Limiter holder terminal temperature.- The temperature of the limiter holder terminals shall be monitored during time-current measurements to assure the limiter holder and attached leads return to the correct ambient temperature between tests.

4.6.14 Alternating current rupture.- The rupture test shall be conducted with the limiter connected as shown on figure 4 for the line-to-neutral tests and as shown on figure 5 for the line-to-line test. These tests shall use the current available from the three phase system specified in 3.5.6. Tests shall be performed in accordance with the following schedule:

Test sample	Altitude	Line-to-neutral	Line-to-line
A	Sea level	X	
B	Sea level	X	
C	60,000 feet 1/	X	
D	60,000 feet 1/	X	
E and F	Sea level		X
G and H	60,000 feet 1/		X

1/ The limiters shall be held at the test altitude and at the ambient temperature of 45° C for 2 hours prior to and during this test. The arcing time shall not be greater than 0.5 cycle or 1/800 second as determined from the oscillograph.

4.6.15 Direct current rupture.- Using the circuitry of figure 4, test the limiter to insure that the requirements of 3.5.7 have been complied with.

4.6.16 Explosion proof test.- The apparatus used for conducting this test shall be in accordance with MIL-E-5272, with provisions for the following tests:

(a) A circulating fan shall be provided to insure adequate mixing of the explosive atmosphere.

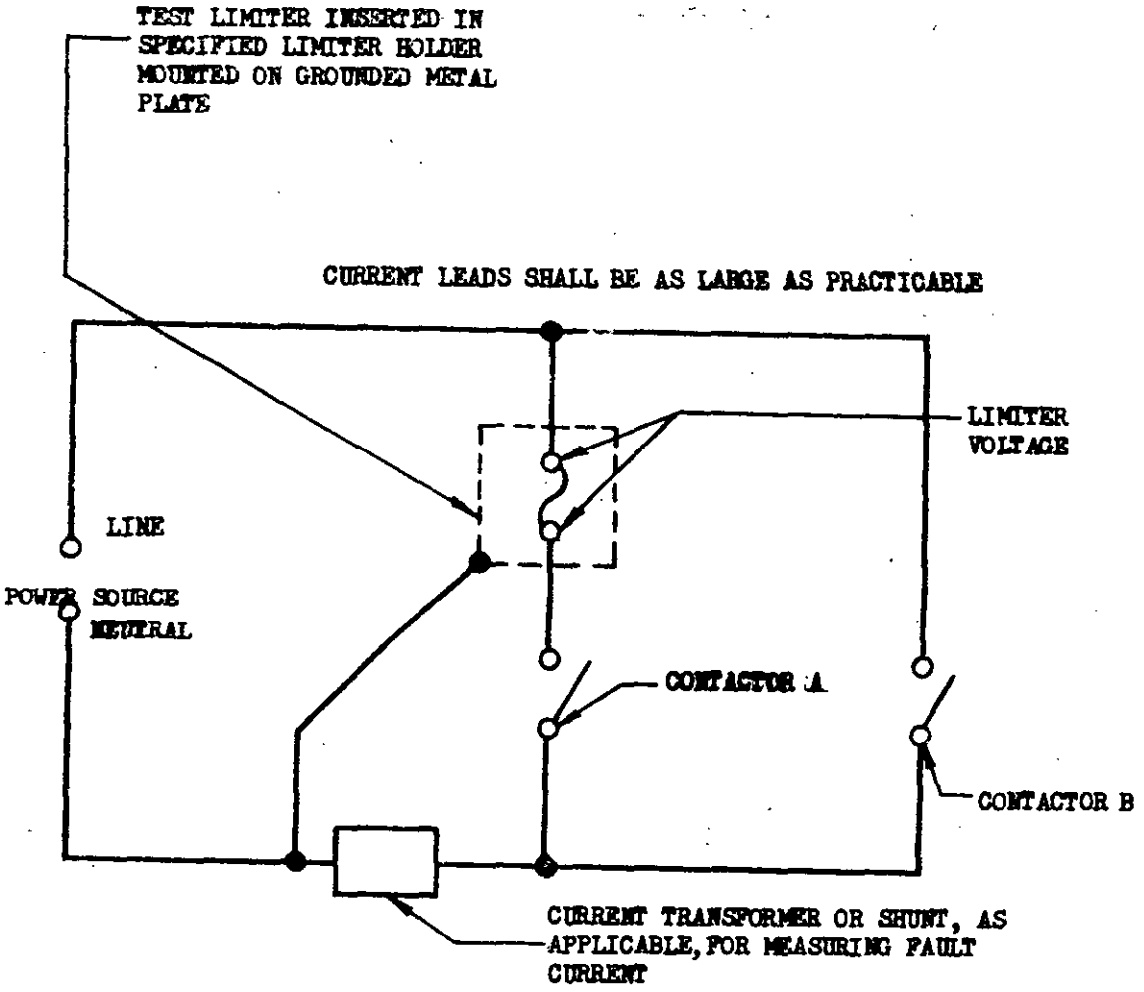
(b) Suitable provisions for heating the explosion chamber shall be provided.

(c) A vacuum pump capable of evacuating the explosion chamber to 0.5 inch of mercury absolute pressure shall be provided.

(d) The limiter shall be inserted in the specified limiter holder and installed inside the explosion chamber.

(e) The explosive atmosphere shall consist of propane gas (C_3H_8) having a purity of 99 percent and certified dry air. The mixing cycle shall be closely controlled to insure repeatability of results.

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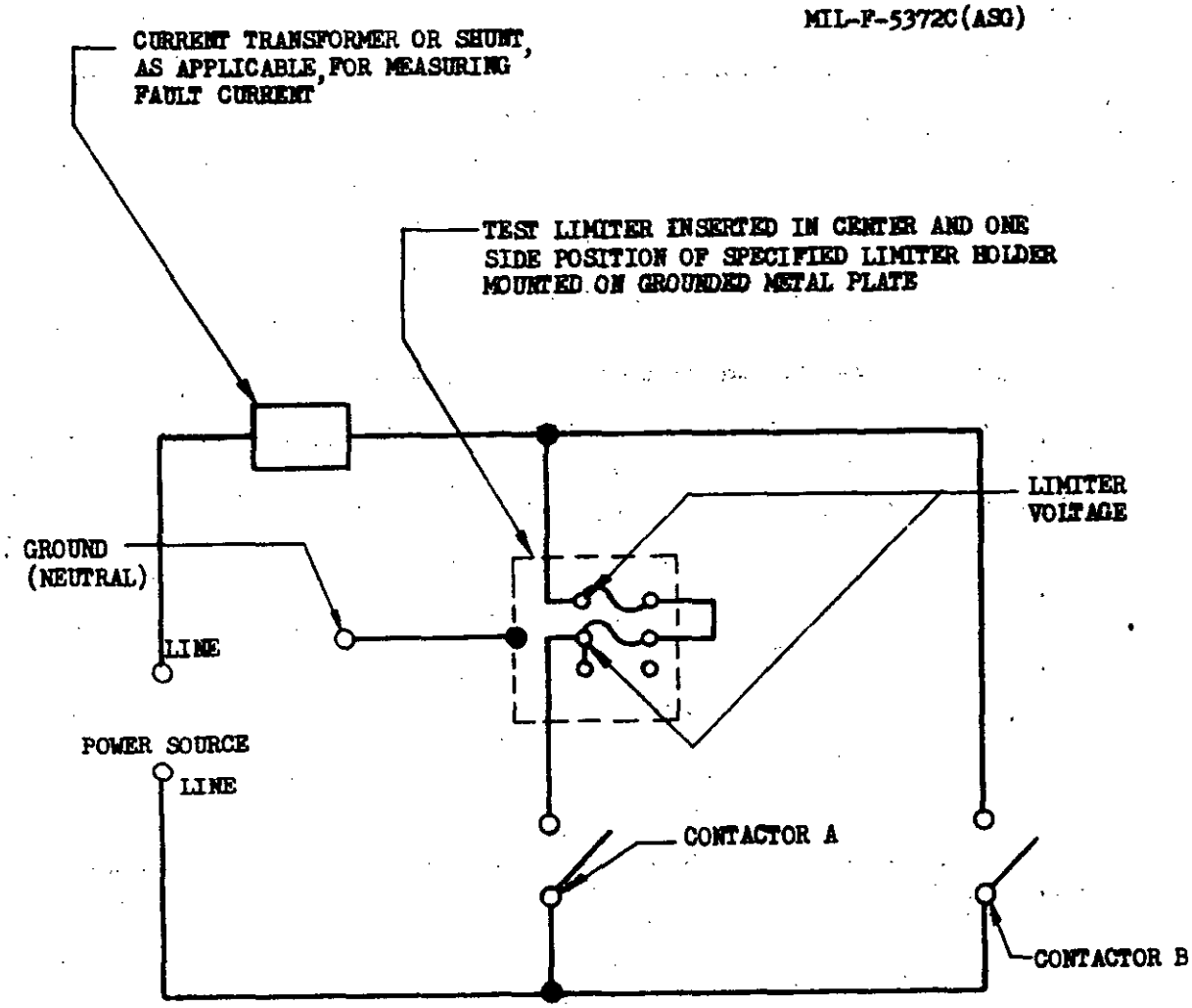
CONTACTOR A SHALL NOT CONTAIN BOUNCE EXCEEDING 250 MICRO-SECONDS. AN OSCILLOGRAPH HAVING A RESPONSE RATING OF AT LEAST 3000 CPS SHALL BE USED TO RECORD TEST CURRENT AND LIMITER VOLTAGES. TIME MEASUREMENTS SHALL BE ACCURATE TO 0.1 MILLISECOND.

PROCEDURE:

(a) WITH CONTACTOR A OPEN, CLOSE CONTACTOR B TO DETERMINE IF THE CURRENT CAPACITY OF THE CIRCUIT IS IN ACCORDANCE WITH THE REQUIREMENT.
(b) OPEN CONTACTOR B.
(c) INSERT TEST LIMITER IN LIMITER HOLDER.
(d) CLOSE CONTACTOR A.

FIGURE 4. Line-to-neutral rupture test circuit

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CURRENT LEADS SHALL BE AS LARGE AS PRACTICABLE.

CONTACTOR A SHALL NOT CONTAIN BOUNCE EXCEEDING 250 MICRO-SECONDS. AN OSCILLOGRAPH HAVING A RESPONSE RATING OF AT LEAST 3000 CPS SHALL BE USED TO RECORD TEST CURRENT AND LIMITER VOLTAGE. TIME MEASUREMENTS SHALL BE ACCURATE TO 0.1 MILLISECOND.

PROCEDURE:

(a) WITH CONTACTOR A OPEN, CLOSE CONTACTOR B TO DETERMINE IF THE CURRENT CAPACITY OF THE CIRCUIT IS IN ACCORDANCE WITH THE REQUIREMENT.
(b) OPEN CONTACTOR B.
(c) INSERT TEST LIMITER IN LIMITER HOLDER.
(d) CLOSE CONTACTOR A.

FIGURE 5. Line-to-line rupture test circuit

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(f) For each explosive mixture or altitude condition, the explosion chamber shall first be evacuated to 0.5 inch of mercury before admission of the explosive mixture.

(g) After the test conditions within the chamber have been achieved and stabilized for the explosive atmosphere, the circulating fan shall be turned off to allow turbulence to subside.

(h) The test currents and circuit used shall be those specified in 3.5.6 for the interruption tests.

(i) The ambient temperature within the chamber shall be 125° ±5° C.

(j) The atmosphere within the chamber shall be verified as explosive after each limiter blowing.

Test sample	Configuration	Altitude	Explosive mixture percent propane by volume
A	Line-to-neutral	Sea level	5.0 percent
B	Line-to-neutral	Sea level	4.5 percent
C	Line-to-neutral	Sea level	5.5 percent
D	Line-to-neutral	15,000 feet	5.0 percent
E	Line-to-neutral	25,000 feet	5.0 percent
F and G	Line-to-line	Sea level	5.0 percent

Failure shall be indicated by ignition of the explosive mixture resulting from limiter blowing.

4.6.17 Sand and dust.- While inserted in the specified limiter holder, the limiter shall be subjected to Procedure I of MIL-E-5272. At the end of this test, the calibration of 500 percent of rated current shall be within the limits required by 4.6.13.

4.6.18 Acceleration.- With the limiter normally inserted in the specified limiter holder, Procedure III of MIL-E-5272 shall be conducted. After this test, the limiter shall meet the requirements of 4.6.13 for time-current tests at 500 percent of rated current.

4.6.19 Mechanical shock.- Three limiters shall be inserted in the specified limiter holder which shall be mounted, by normal means, on a mounting fixture and subjected to a shock test consisting of 30 shocks (10 shocks in each of 3 mutually perpendicular planes, 5 each at 6 directions) each of 25g intensity. The methods and apparatus utilized for the procedure of the shock pulses shall be in accordance with MIL-STD-202. The shock and pulse duration shall be monitored on the mounting fixture in the proximity of the support point of the limiter holder. The limiter shall remain inserted in the limiter holder throughout the test.

4.6.20 Terminal voltage drop.- The limiter shall be inserted in the specified limiter holder and the voltage drop shall be determined with 100 percent rated current applied.

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4.6.21 Temperature rise.- Three limiters shall be inserted in the specified limiter holder, with leads attached suitable for calibration, and be subjected to the following duty cycle. This test shall be repeated for a total of 25 duty cycles.

Percent load	Time in minutes
100	30
140	1
0	29

The maximum temperature at any given point on the assembly shall not be greater than allowed in 3.5.8, with ambient temperature constant throughout the test.

4.6.22 Rated current.- Subsequent to certain of the foregoing tests, as specified in table I, 100 percent rated current shall be applied with the limiter in the applicable holder.

4.6.23 Packaging, packing, and marking.- Preparation for delivery shall be examined for conformance to section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking.- Unless otherwise specified, fuses shall be preserved and packaged in accordance with Method III, level A or C of MIL-P-17555 in unit quantities of one each. They shall be packed in accordance with level A, B, or C, as specified (see 6.2), and marked in accordance with MIL-P-17555. Unless otherwise specified, shipment marking nomenclature shall be as follows:

FUSE, CURRENT LIMITER TYPE, AIRCRAFT
MS part number as applicable

6. NOTES

6.1 Intended use.- The current limiters covered by this specification are intended for use in aircraft primary electrical power systems with an alternating current rating 115/200V at 400 cps and a direct current rating of 28V. The main purposes of these devices are to protect distribution wiring from excessive current flow and to prevent collapse of the electrical system due to abnormal fault conditions, to be usefully applicable in coordinated electric systems.

6.2 Ordering data.- Procurement documents should specify:

(a) Title, number, and date of this specification.
(b) Quantity and ampere rating conforming to the applicable MS (see 1.2).
(c) Levels of preservation, packaging, and packing (see section 5).
(d) If additional data is required (see 3.2).

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6.3 Qualification.- With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set of opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Bureau of Naval Weapons, Department of the Navy, Washington, D. C., 20360; however, information pertaining to qualification of products may be obtained from: Commanding Officer, U. S. Naval Air Development Center, Johnsville, Pa.

6.4 Definitions.- The following definitions apply to terminology used in this specification.

6.4.1 Arcing time.- Arcing time is the time measured from that point when element melt time ends to that point when current is interrupted and permanently becomes zero. If a mechanical indicator is utilized which incorporates a secondary element parallel to the fusible element, arcing time will commence from the point at which indicator melt time ends.

6.4.2 Current droop.- Current droop is the percentage decrease of the fusing current from the moment fusing current is initiated to the point where indicator melt time or arcing time begins. Current droop is usually caused by increase in limiter element resistance due to increased temperature caused by flow of current. Current droop can be encountered if care is not exercised in the selection of interconnecting wire, resistive loads, and battery source. (See figure 3.)

$$\text{Current droop} = \frac{\text{Initial current} - \text{final current}}{\text{initial current}} \times 100$$

6.4.3 Element melt time.- Element melt time is the time elapsed from the moment a fusing current beings to flow to the moment the current sharply drops in value and indicator melt time or arcing commences.

6.4.4 Current limiter.- A current limiter is an aircraft fuse designed specifically with a high temperature melting point to provide protection for electrical power distribution systems against fault current.

6.4.5 Limiter current rating.- Limiter current rating is a value of current assigned to a limiter that is below the minimum fusing current and can be carried continuously under all environmental conditions encountered by the limiter without deterioration of the element, the body, or the terminals.

6.4.6 Limiter element.- That part of a limiter designed to melt by an excessive flow of current and thus interrupt a faulty condition is a limiter element.

6.4.7 Limiter voltage rating.- Limiter voltage rating is that value assigned to a limiter denoting the maximum open-circuit voltage it can safely interrupt under specified test conditions.

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6.4.8 Limiter holder.- A limiter holder is a mounting device with contacts and terminals for the purpose of accepting limiters for easy connection within a circuit.

6.4.9 Fusing current.- A fusing current is a current of such a magnitude that, under specified environmental conditions, the limiter element will melt.

6.4.10 Indicator melt time.- The lapsed time from the moment element melt time ends and arcing commences is indicator melt time. (See figure 3.)

6.4.11 Interrupt capacity.- The largest prospective current the fuse can safely interrupt under specified conditions of voltage and of power factor or time constant is interrupt capacity.

6.4.12 Minimum fusing current.- The smallest value of current that will melt the limiter element at a specified ambient temperature is the minimum fusing current.

6.4.13 Prospective current.- That value of short-circuit current (rms or dc) that would flow in a circuit if the limiter element were replaced by an element of negligible impedance is the prospective current.

6.4.14 Self-indicating current limiter.- That type of limiter that incorporates a device to visually denote severance of the fusible element as an integral part of the limiter is a self-indicating current limiter. Those limiters that visually display the fusible element or incorporate a mechanical indicating device are classified as self-indicating current limiters.

6.4.15 Fuse melt time.- Fuse melt time is the total time obtained by adding element melt time, indicator melt time, and arcing time (see figure 2).

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use of the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

MIL-F-5372C(ASG) Fuse, Current Limiter Type, Aircraft

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ Direct Government Contract ☐ Subcontract

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS OF ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)

SUBMITTED BY (Printed or typed name and activity)	DATE

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